INCH-POUND

The documentation and process conversion measures necessary to comply with this revision shall be completed by: 21 October 2022.

MIL-PRF-19500/427T
21 June 2022
SUPERSEDING
MIL-PRF-19500/427R
w/AMENDMENT 2
9 December 2020
(See 6.6)

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER,
ENCAPSULATED (THROUGH-HOLE AND SURFACE MOUNT PACKAGES), AND UN-ENCAPSULATED,
TYPES 1N5614, 1N5616, 1N5618, 1N5620, 1N5622,
QUALITY LEVELS JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, hermetically sealed power rectifier diodes. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each encapsulated device types as specified in MIL-PRF-19500. Two levels of product assurance (JANHC and JANKC) are provided for each unencapsulated device type.

1.2 Package outlines and die topography. The device package for the encapsulated device type are as follows: Axial leaded in accordance with figure 1, square end-cap (US) surface mount version in accordance with figure 2, and planar ceramic (UMC) surface mount version in accordance with figure 3. The characteristics, dimensions and topography for JANHC and JANKC un-encapsulated die are as follows: A version die in accordance with figure 4, and B version die in accordance with figure 5.

1.3 Maximum ratings. Unless otherwise specified T_A = 25°C and ratings apply to all package outlines.

1.3.1 Ratings applicable to all types. T_{STG} = $T_{J(max)}$ = -65°C to +175°C.

1.3.2 Ratings applicable to individual types.

Col. 1	Col. 2	Col. 3	Col. 4	Col. 5	Col. 6	Col. 7	Col. 8	Col. 9	Col. 10
Types (1)	V_{RWM}	I_{O1} T_A = +55°C (2)	I_{O2} T_A = +100°C (3)	I_{FSM} T_A = +55°C I_F = 750 mA t_P = 8.3 ms	t_{rr}	$R_{\theta JL}$ (4) (5)	$R_{\theta JEC}$ (4) (5) (6)	$R_{\theta JX}$ (4)	$R_{\theta JSP}$ (7)
	V	A	mA	A	µs	°C/W	°C/W	°C/W	°C/W
1N5614, US, UMC	200	1	750	30	2	36	13	133	22.4
1N5616, US	400	1	750	30	2	36	13	133	
1N5618, US	600	1	750	30	2	36	13	133	
1N5620, US	800	1	750	30	2	36	13	133	
1N5622, US	1,000	1	750	30	2	36	13	133	

See notes on next page.

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1.3.2 Ratings applicable to individual types − Continued.

(1) Barometric pressure: 1N5616 and 1N5618 = 8 mm Hg, 1N5620 and 1N5622 = 33 mm Hg.
(2) Derate linearly at 5.56 mA/°C from 1 A at T_A = +55°C, to 0.75 A at T_A = +100°C.
(3) Derate linearly at 10 mA/°C from 0.75 A at T_A = +100°C, to 0 A at T_A = +175°C.
(4) For the 1 A rating at 55°C ambient and the 750 mA rating at 100°C ambient, these I_O ratings are for a thermal mounting methods (PC boards or other) where the lead or end-cap temperatures cannot be maintained as shown in col. 3 and where thermal resistance from mounting point to ambient is still sufficiently controlled where $T_{J(MAX)}$ in 1.3.1 is not exceeded. This equates to $R_{\theta JX} \leq 133$°C/W as shown in col. 9. Also see application notes in 6.7.
(5) T_L at L = .375 inch (9.52 mm) for axial leaded devices. $T_{EC} = T_L$ at L = 0 inch (0 mm) for US suffix devices.
(6) US suffix devices only.
(7) See 6.7 for solder pad size calculation for 1N5614UMC only.

1.4 Primary electrical characteristics. Primary electrical characteristics are not applicable to this specification sheet.

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level.

1.5.1.1 Quality level designators for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV", and "JANS".

1.5.1.2 Quality level designators for unencapsulated devices (die). The quality level designators for unencapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

1.5.2 Device type. The designation system for the device types of semiconductors covered by this specification sheet are as follows.

1.5.2.1 First number and first letter symbols. The semiconductors of this specification sheet use the first number and letter symbols "1N".

1.5.2.2 Second number symbols. The second number symbols for the semiconductors covered by this specification sheet are as follows: "5614", "5616", "5618", 5620", and "5622".

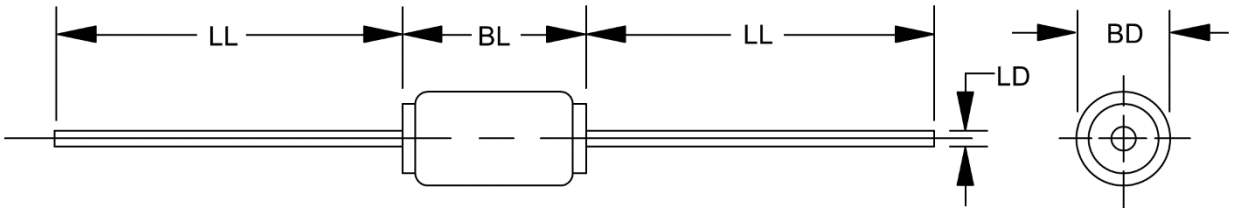
1.5.3 Suffix symbols. The following suffix symbols are incorporated in the PIN as applicable.

	A blank suffix symbol indicates a through-hole mount axial leaded package (see figure 1).
US	Indicates a square end-cap surface mount package (see figure 2).
UMC	Indicates a planar ceramic surface mount package (1N5614 only, see figure 3).

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

1.5.5 Die identifiers for unencapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifiers that are applicable for this specification sheet are "A" and "B" (see figures 4 and 5 and 6.5.2).

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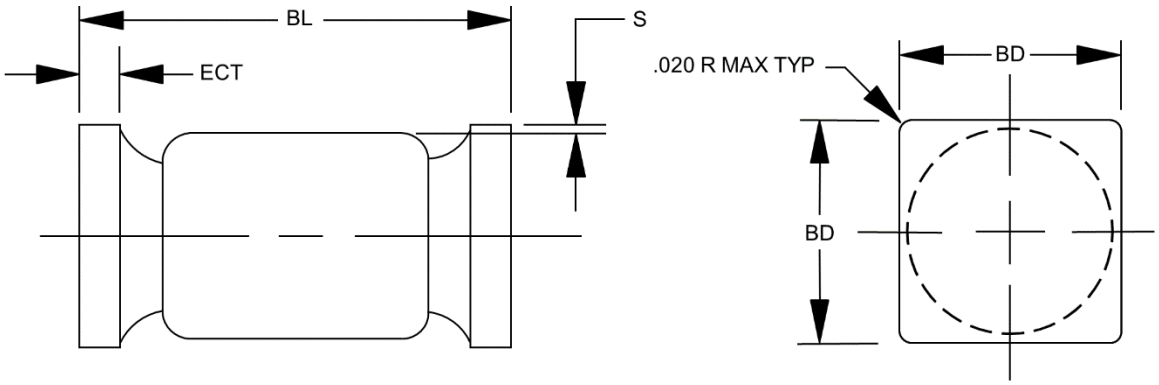


Symbol	Incxhes	Inches	mm	mm	Notes
	Min	Max	Min	Max	
BD	.065	.110	1.65	2.79	3
BL	.130	.205	3.30	5.21	4
LD	.026	.033	0.66	0.84	
LL	1.00	1.50	25.4	38.10	

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Dimensions are pre-solder dip.
3. Dimension BD shall be measured at the largest diameter.
4. The BL dimension shall include the entire body including plugs and sections of the leads over which the diameter is uncontrolled. This uncontrolled area is defined as the zone between the edge of the diode body and extending .050 inch (1.27 mm) onto the leads.
5. The shape of the body, within the bounds of the dimensions is optional.
6. In accordance with ASME Y14.5, diameters are equivalent to øx symbology.

FIGURE 1. Physical dimensions for axial leaded devices (for non-US suffix devices only).

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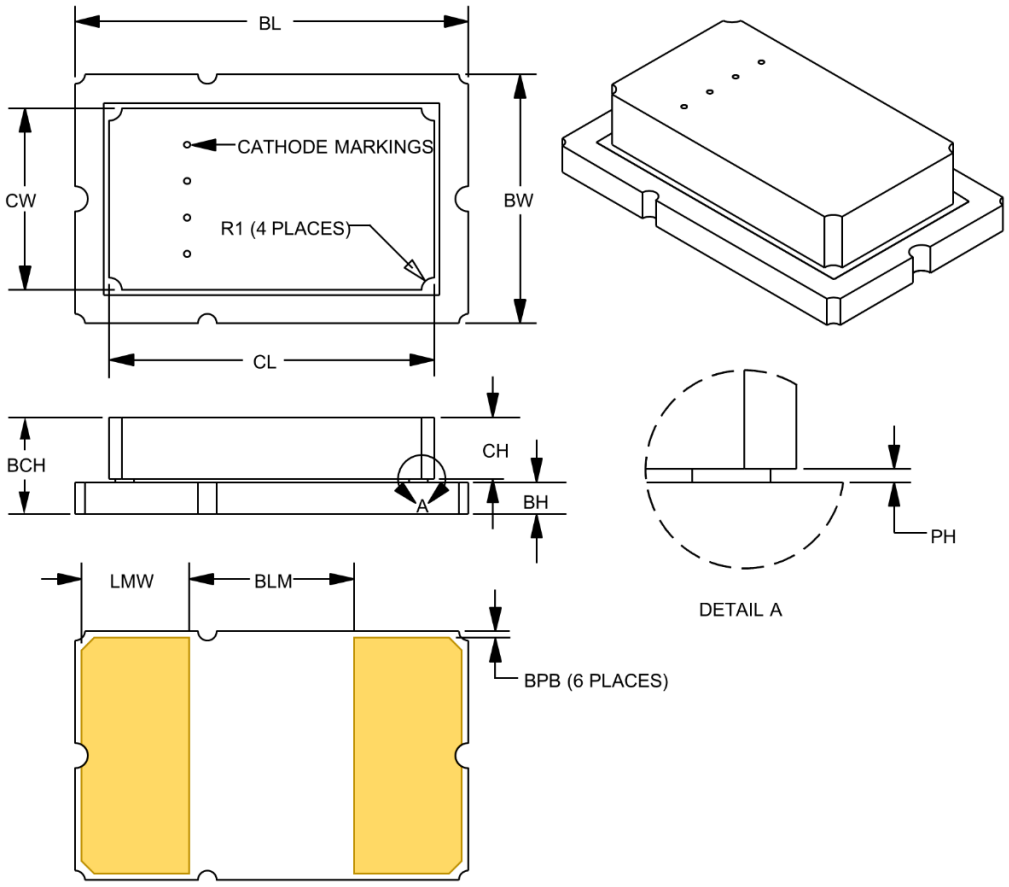


Symbol	Inches	Inches	mm	mm
	Min	Max	Min	Max
BD	.091	.103	2.31	2.62
BL	.168	.200	4.27	5.08
ECT	.019	.028	0.48	0.71
S	.003		0.08	

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Dimensions are pre-solder dip.
3. The S dimension is minimum clearance of glass body to mounting surface on all orientations.
4. In accordance with ASME Y14.5, diameters are equivalent to øx symbology.

FIGURE 2. Physical dimensions for square end-cap surface mount (US) package.

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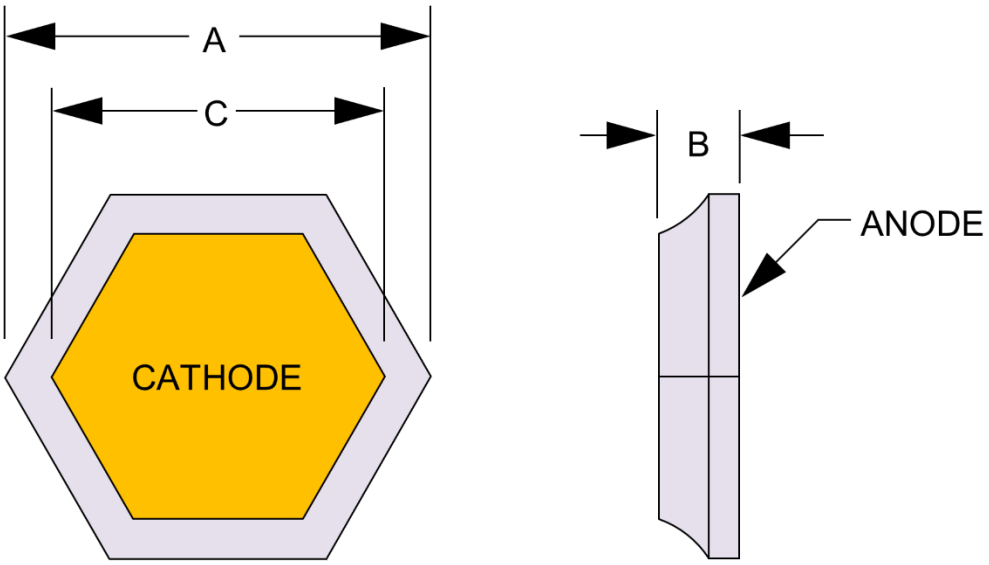


Symbol	Inches	Inches	mm	mm		Symbol	Inches	Inches	mm	mm
	Min	Max	Min	Max			Min	Max	Min	Max
BCH	.058	.064	1.46	1.63		CH	.038	.040	0.98	1.01
BH	.018	.022	0.46	0.56		CL	.199	.211	5.06	5.36
BL	.242	.254	6.15	6.45		CW	.109	.120	2.76	3.06
BLM	.098	.110	2.49	2.79		LMW	.062	.074	1.58	1.88
BW	.151	.163	3.84	4.14		PH	.001	.003	0.03	0.07
BPB	.003 Nominal		0.08 Nominal			R1	.006 Nominal		0.15 Nominal	

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Shaded areas denote metallized areas.
3. In accordance with ASME Y14.5, diameters are equivalent to øx symbology.

FIGURE 3. Physical dimensions of planar ceramic (UMC) surface mount package (1N5614 only).

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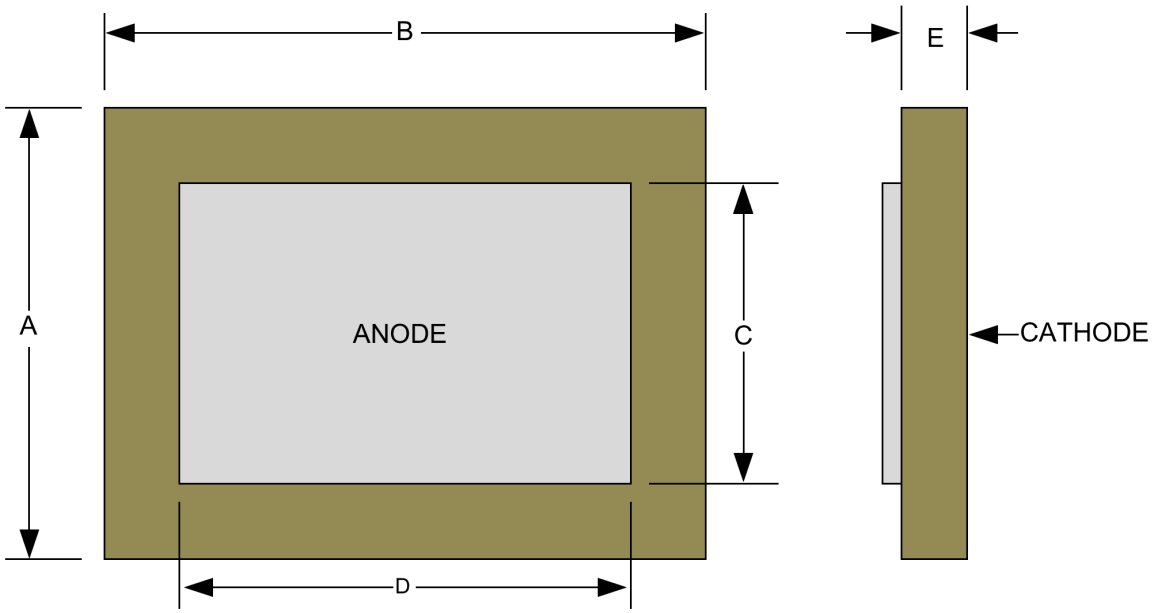
A - version

Letter	Inches	Inches	mm	mm
	Min	Max	Min	Max
A	.047	.053	1.19	1.35
B	.007	.011	0.18	0.28
C	.033	.037	0.84	0.94

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. The metallization characteristics of the die are:
   Top metal: Gold 10,000 Å minimum.
   Back metal: Gold 4,000 Å minimum.

FIGURE 4. Physical dimension, JANHCA and JANKCA die.

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B - version

Letter	Inches	Inches	mm	mm
	Min	Max	Min	Max
A	.040	.050	1.016	1.270
B	.055	.065	1.397	1.651
C	.024	.034	0.610	0.864
D	.039	.049	0.991	1.245
E	.003	.013	0.076	0.330

NOTES:
1. Dimensions are in inches. 2. Millimeters are given for general information only.
2. The metallization characteristics of the die are:
   Top metal (anode) Aluminum at 30,000 Å ± 5,000 Å.
   Back metal (cathode) Gold at 6,500 Å ± 1,500 Å.

FIGURE 5. Physical dimension, JANHCB and JANKCB die.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods For Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

EC	End-cap.
P_{RM}	Peak reverse power.
US	Short-body unleaded or surface mount diodes (square end-caps).
UMC	Planar ceramic surface mount package with ceramic lid.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1, 2, 3, 4, and 5 herein.

3.4.1 Lead finish. Unless otherwise specified, lead, end-cap or pad finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. When solder alloy is used for finish, the maximum lead temperature is limited to 175°C maximum. Where a choice of finish is desired, it shall be specified in the acquisition document (see 6.2).

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3.4.2 Diode construction.

3.4.2.1 Axial leaded. Round body axial leaded devices shall be constructed utilizing non-cavity double plug construction with high temperature metallurgical bonding between both sides of the silicon die and terminal pins. Metallurgical bonds shall be in accordance with the requirements of category I of MIL-PRF-19500. Point contact construction shall not be acceptable. Silver button Dumet design shall not be acceptable.

3.4.2.2 Surface mount. US version devices shall be structurally identical to the round body axial leaded version devices except for lead termination. The die in the UMC version devices shall be eutectically mounted and wire bonded in a ceramic package.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5.1 Marking of US version devices. For US version devices only, all marking (except for polarity) may be omitted from the body, but shall be retained on the initial container.

3.5.2 Polarity. The polarity of all types shall be indicated with a contrasting color band to denote the cathode end. Alternatively, for US version devices, a minimum of three contrasting color dots spaced around the periphery on the cathode end or a contrasting color band may be used. No color coding will be permitted. For UMC version devices polarity shall consist as a minimum of 4 contrasting color dots marked on the lid.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3 and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I, herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

d. Element evaluation of un-encapsulated die (see 4.6).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E inspection. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.2 JANHC and JANKC die. Qualification shall be in accordance with appendix G of MIL-PRF-19500 and as specified herein.

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4.3 Screening (quality levels JANTX, JANTXV and JANS only). Screening of packaged devices shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screening	Measurements for JANS level	Measurements for JANTXV and JANTX level
3a (1) 3c	Required Thermal impedance (see 4.3.1).	Required Thermal impedance (see 4.3.1).
5	Applicable to UMC devices only.	Not required
9	I_{R1} and V_{F2}	Not required
10	Method 1038 of MIL-STD-750, condition A	Method 1038 of MIL-STD-750, condition A
(2) 11	I_{R1} and V_{F2}, Δ_{IR1} ≤ 100 percent of initial reading or ±100 nA dc, whichever is greater. ΔV_{F2} ≤ ±0.1 V dc	I_{R1} and V_{F2}
12	Required, see 4.3.2	Required, see 4.3.2
(2) (3) 13	Subgroups 2 and 3 of table I herein: ΔI_{R1} ≤ 100 percent of initial reading or ±100 nA dc, whichever is greater. ΔV_{F2} ≤ ±0.1 V dc. Scope display evaluation (see 4.5.3)	Subgroup 2 of table I herein: ΔI_{R1} ≤ 100 percent of initial reading or ±100 nA dc, whichever is greater. ΔV_{F2} ≤ ±0.1 V dc. Scope display evaluation (see 4.5.3)

(1) Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with MIL-PRF-19500, screen 3 prior to this thermal test.
(2) For JANTX and JANTXV devices, ΔV_{F2} may be omitted if thermal impedance is performed, unless irradiation is used to reduce the carrier lifetime.
(3) Thermal impedance is not required in screen 13, if already previously performed.

4.3.1 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3101 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, and K factor where appropriate. The limits will be statistically derived. See table E-IX of MIL-PRF-19500, group E, and table II, subgroup 4 herein.

4.3.2 Power burn-in conditions. Power burn-in shall be performed in accordance with test condition B of method 1038 of MIL-STD-750. The details of 4.5.2 and those that follows shall apply: $I_{O(min)}$ = 1 A minimum. Adjust I_O or T_A to achieve the required T_J. T_J = 135°C minimum and 175°C maximum. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, T_J, mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein. The $Z_{\theta JX}$ end-point shall be derived by the supplier and approved by the qualifying activity. The $Z_{\theta JX}$ end-point shall also be documented in the qualification report.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and herein. Delta measurements shall be as specified in table III herein.

4.4.2.1 Quality level JANS, table E-VIA of MIL-PRF-19500.

Subgroup	Method	Conditions
B3	4066	Condition A, I_{FSM} = rated I_{FSM} (see col. 5 of 1.3.2); ten surges of 8.3 ms each at 1 minute intervals, superimposed on I_O = 1 A, V_{RWM} = rated V_{RWM} (see col. 2 of 1.3.2), T_A = +55°C maximum.
B4	1037	I_O = 1 A minimum for applicable mounting and required delta temperature in method 1037 of MIL-STD-750; V_R = rated V_{RWM} (see col. 2 of 1.3.2); 2,000 cycles.
B5	1027	See 4.5.2. I_O = 1 A minimum; apply V_R = rated V_{RWM} (see col. 2 of 1.3.2) adjust I_O or T_A to achieve T_J = +175°C minimum. f = 50 to 60 Hz; t = 1,000 hours. n = 45, c = 0. Temporary leads may be added for surface mount devices.
B8	4065	P_{RM} ≥ 318 W. Test shall be performed on each sublot; sampling plan n = 10, c = 0, electrical end-points, see table I, subgroup 2 and table III herein.

4.4.2.2 Quality levels JAN, JANTX and JANTXV, table E-VIB of MIL-PRF-19500.

Subgroup	Method	Conditions
B3	1027	See 4.5.2; I_O = 1A minimum (see col. 3 of 1.3.2). Adjust I_O or T_A to achieve the required T_J of +150°C minimum. Apply V_R = rated V_{RWM} (see col. 2 of 1.3.2), f = 50 to 60 Hz. Temporary leads may be added for surface mount devices.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows. Delta measurements shall be as specified in table III herein. Leaded samples may be used in lieu of surface mount devices (US version) for life tests.

Subgroup	Method	Condition
C2	2036	Axial leaded devices only. Tension: Test condition A; weight = 12 pounds (5.44 Kg); t = 15 s. Lead fatigue: Test condition E; weight = 2 pounds (0.91 Kg).
C2	2038	US surface mount devices only. Tab pull: Test condition B; weight = 12 pounds (5.44 Kg); t = 15 s. NOTE: Tab pull is not applicable for axial leaded devices. Test method 2038 not required for UMC devices.
C5	4081	See 4.3.1. For axial leaded devices, $R_{\theta JL}$ (maximum) ≤ 36°C/W, L= .375 inch (9.53 mm) . For US surface mount devices, $R_{\theta JEC}$ ≤ 13°C/W. For UMC surface mount devices, $R_{\theta JA}$ see col. 9 and $R_{\theta JSP}$ see col. 10 of 1.3.2 herein.
C6	1026	See 4.5.2; I_O = 1 A minimum (see col. 3 of 1.3.2); adjust I_O or T_A to achieve T_J = +150°C minimum. Apply V_R = rated V_{RWM} (see col. 2 of 1.3.2), f = 50 to 60 Hz.

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4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein; except, $Z_{\theta JX}$ need not be performed. See table III herein for delta limits when applicable.

4.5 Methods of inspection. Methods of inspection shall be specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Free air burn-in and life tests. These tests shall be conducted with a half-sine waveform of the specified peak voltage impressed across the DUT in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall be neither greater than 180 degrees, nor less than 150 degrees. The use of a current limiting or ballast resistor is permitted provided that each DUT still sees the full required T_J and I_O and that the minimum required voltage, where applicable, is maintained through out the burn-in period. Test method 3100 of MIL-STD-750 shall be used to measure T_J. T_J shall be 135°C minimum for screening and 150°C minimum for life tests.

4.5.3 Scope display evaluation. Scope display evaluation shall be performed using condition B (sharp and stable) in accordance with method 4023 of MIL-STD-750. Scope display may be performed on ATE (automatic test equipment) for screening only with the approval of the qualifying activity. Scope display in table I, subgroup 4 herein shall be performed on a scope. Reverse current (I_{BR}) over the knee shall be 500 µA peak.

4.5.4 Thermal resistance. Thermal resistance measurement shall be performed in accordance with method 4081 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H and t_{MD}. See table E-IX of MIL-PRF-19500, subgroup 4, and figures 6, 7, and 8 herein. Forced moving air or draft shall not be permitted across the devices during test.

4.5.5 Peak reverse power characterization. The testing to characterize P_{RM} shall be conducted in accordance with method 4065 of MIL-STD-750. The following details shall apply:

a. During the test, the voltage (V_{BR}) shall be monitored to verify it has not collapsed.

b. Any collapse in V_{BR} during or after the test, or rise in leakage current (I_R) after the test that exceeds I_{R1} in table I, shall be considered a failure to that level of applied P_{RM}.

c. Progressively higher levels of P_{RM} shall be applied until failure occurs on all devices within the chosen sample size to characterize each sublot.

4.6 Element evaluation of un-encapsulated die. The element evaluation of un-encapsulated die shall be in accordance with appendix G of MIL-PRF-19500. As a minimum, die shall be 100-percent probed to ensure compliance with table I, subgroup 2 of MIL-PRF-19500. Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance 2/	3101	See 4.3.1	$Z_{\theta JX}$			°C/W
Forward voltage	4011	Condition B, I_F = 1 A dc (pulsed, see 4.5.1); tp ≤ 8.3 ms, 2 percent maximum duty cycle	V_{F1}	0.6	1.1	V
Forward voltage	4011	Condition B, I_F = 3 A dc pulsed (see 4.5.1) tp ≤ 8.3 ms, 2 percent maximum duty cycle	V_{F2}	0.8	1.3	V
Reverse current leakage	4016	Condition A or B, V_R = rated V_{RWM} (see col. 2 of 1.3.2)	I_{R1}		0.5	µA dc
Breakdown voltage	4021	I_R = 50 µA dc	V_{BR}			
1N5614, US, UMC				220		V
1N5616, US				440		V
1N5618, US				660		V
1N5620, US				880		V
1N5622, US				1,100		V
Subgroup 3						
High temperature operation:		T_A = +125°C				
Reverse current leakage	4016	Condition A or B, V_R = rated V_{RWM} (see col. 2 of 1.3.2)	I_{R2}		25	µA dc
Low temperature operation		TA = -55°C				
Forward voltage	4011	Condition B; I_F = 3 A, pulsed (see 4.5.1), t_P ≤ 8.3 ms; 2 percent maximum duty cycle	V_{F3}		1.5	V

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Reverse recovery time	4031	Condition B1	t_{rr}		2	µs
Scope display	4023	See 4.5.3; n = 116, c = 0				
Subgroup 5						
Not applicable						
Subgroup 6						
Forward surge current	4066	Condition A, I_{FSM} = 30 A (pk); ten surges of 8.3 ms each at 1 minute intervals, superimposed on I_O = 1 A, V_{RWM} = rated V_{RWM} (see col. 2 of 1.3.2); T_A = +55°C				
Electrical end-points		See table I, subgroup 2				
Subgroup 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.
2/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroups 3 and 5 (JANS).
Group B, subgroups 2 and 3 (JAN, JANTX, and JANTXV).
Group C, subgroup 2 and 6.
Group E, subgroup 1A and 1B.

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TABLE II. Group E inspection (all quality levels) for qualification and requalification only.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1A			45 devices, c = 0
Temperature cycling (air to air) 1/	1051	20 cycles, except high temperature shall be 150°C and low temperature shall be -195°C. (Not applicable for UMC package.)	
Hermetic seal	1071	Fine and gross. Fine leak is applicable for UMC version devices only.	
Electrical measurement		See table I, subgroup 2 and table III, steps 1 and 2, herein.	
Subgroup 1B			45 devices, c = 0
Temperature cycling (air to air)	1051	500 cycles, condition C, -65°C to +175°C.	
Hermetic seal 2/	1071		
Electrical measurement		See table I, subgroup 2 and table III, steps 1 and 2, herein.	
Subgroup 2			22 devices, c = 0
Steady state dc blocking life	1048	T_A = +150°C; t = 1,000 hours +65, -0 hours; dc = 80 to 85 percent rated V_{RWM} (see col. 2 of 1.3.2).	
Electrical measurement		See table I, subgroup 2 and table III, steps 1 and 2, herein.	
Subgroup 4			Sample size N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 5			22 devices, c = 0
Barometric pressure (reduced)	1001	Pressure shall be as follows: Condition D (8 mm Hg) for 1N5616, US, and 1N5618, US. Condition C (33 mm Hg) for 1N5620, US, and 1N5622, US.	

See footnotes at end of table.

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TABLE II. Group E inspection (all quality levels) for qualification and requalification only - Continued.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 8			n = 45 devices
Peak reverse power	4065	P_{RM} shall be characterized by the supplier and the data shall be available to the Government. The test to destruction shall be performed on each sublot.	
Electrical measurement		See 4.5.5.	
Subgroup 9			n = 45 devices
Resistance to glass cracking	1057	Test condition B. Step stress to destruction by increasing cycles or up to a maximum of 25 cycles. Not required for UMC package.	
Subgroup 10			22 devices, c = 0
Forward surge current	4066	Condition A, I_{FSM} = rated I_{FSM} (see col. 5 of 1.3.2); ten surges of 8.3 ms each at 1 minute intervals superimposed on I_O = 1.0 A (see col. 3 of 1.3.2); V_{RWM} = rated V_{RWM} (see col. 2 of 1.3.2); T_A = + 55°C.	
Electrical measurement		See table I, subgroup 2 and table III, steps 1 and 2, herein.	

1/ Test method 1056 of MIL-STD-750, test condition D, using liquid nitrogen, may be used in lieu of test method 1051 of MIL-STD-750.
2/ For opaque glass diodes use test method 2068 of MIL-STD-750.

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TABLE III. Groups B and C delta measurements. 1/ 2/ 3/ 4/

Step	Inspection	MIL-STD-750		Symbol	Limits
		Method	Conditions		
1.	Reverse current leakage change	4016	Condition A or B	ΔI_{R1} 5/	For JAN, JANTX, and JANTXV ≤ 250 nA or 100 percent, whichever is greater, For JANS ≤ 100 nA or 100 percent, whichever is greater.
2.	Forward voltage change	4011	Condition B, I_F = 3 A; pulsed (see 4.5.1)	ΔV_{F2} 5/	±50 mV maximum change from previous measured value.

1/ The delta measurements taken during group B inspections for quality level JANS (table E-VIA of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroup 3, the measurements of step 2 of this table shall also be taken.
   b. In addition to the measurements specified for subgroup 4, the measurements of step 2 of this table shall also be taken.
   c. In addition to the measurements specified for subgroup 5, the measurements of steps 1 and 2 of this table shall also be taken.

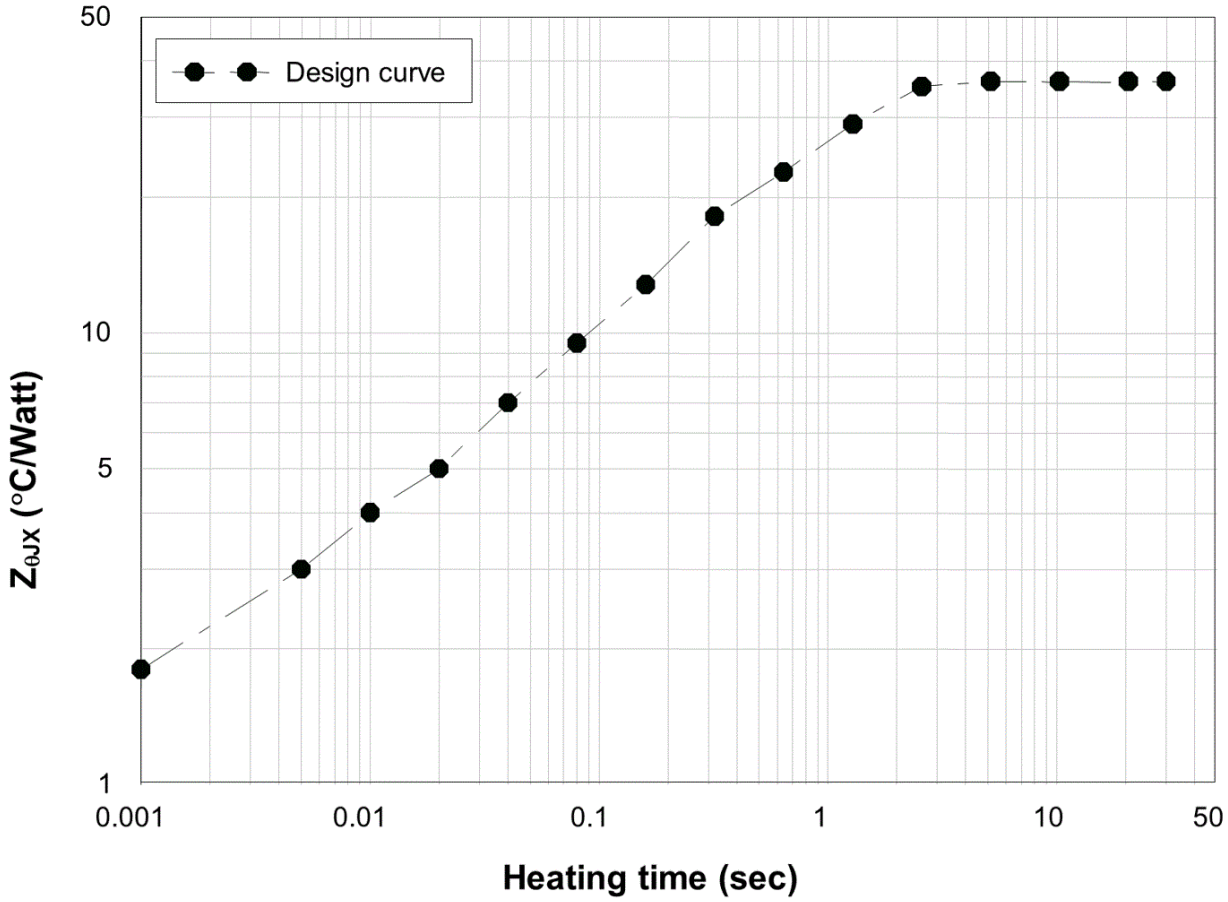
2/ The delta measurements taken during group B inspections for quality levels JAN, JANTX and JANTXV (table E-VIB of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroup 3, the measurements of step 1 of this table shall also be taken.
   b. In addition to the measurements specified for subgroup 6, the measurements of step 1 of this table shall also be taken.

3/ The delta measurements taken during group C inspections for all quality levels (table E-VII of MIL-PRF-19500) shall be as follows:
   a. For quality level JANS, in addition to the measurements specified for subgroup 2, the measurements of steps 1 and 2 of this table shall also be taken.
   b. For quality levels JANS, in addition to the measurements specified for subgroup 6, the measurements of steps 1 and 2 of this table shall also be taken.
   c. For quality levels JAN, JANTX, and JANTXV, in addition to the measurements specified for subgroup 6, the measurements of step 1 of this table shall also be taken.

4/ The delta measurements for taken during group E inspections for all quality levels (table E-IX of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroup 1, the measurements of steps 1 and 2 of this table shall also be taken.
   b. In addition to the measurements specified for subgroup 2, the measurements of steps 1 and 2 of this table shall also be taken.

5/ Devices which exceed the limits of table I, subgroup 2 herein for this test shall not be accepted.

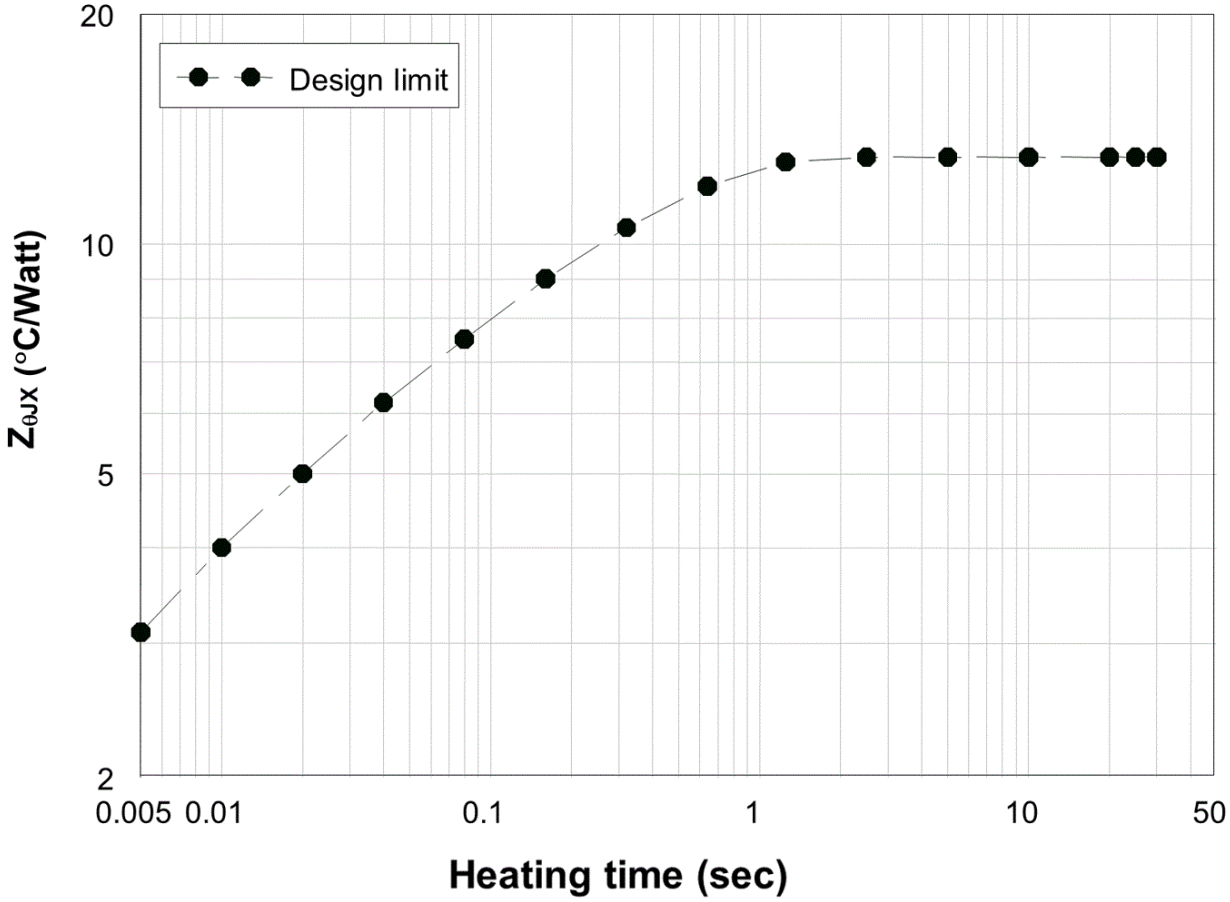
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$R_{\theta JL}$ = 36°C/W

FIGURE 6. Thermal impedance curve ($R_{\theta JL}$) for axial leaded package.

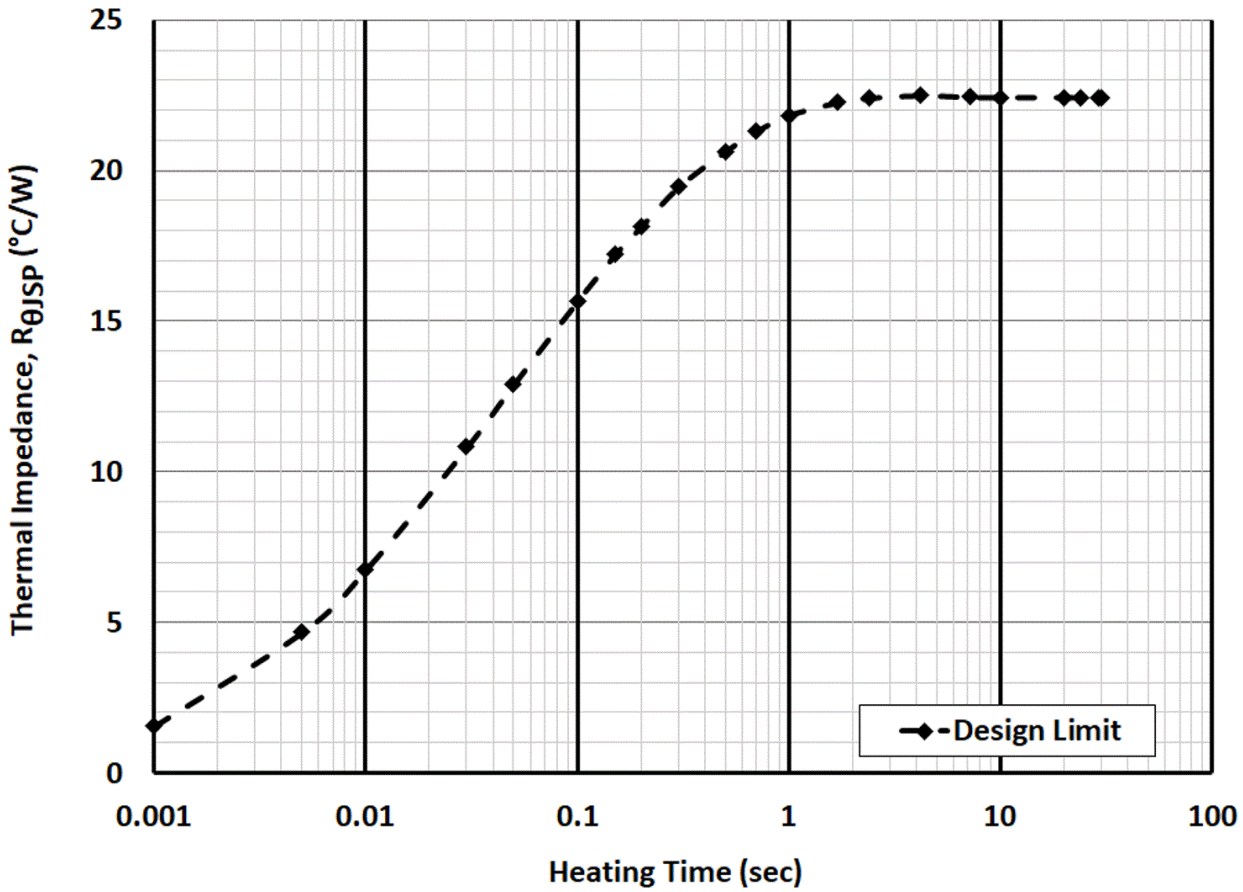
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$R_{\theta JEC}$ = 13°C/W

FIGURE 7. Thermal impedance curve ($R_{\theta JEC}$) for US surface mount package.

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$R_{\theta JSP}$ = 22.4 °C/W for 1N5614UMC

FIGURE 8. Thermal impedance curve ($R_{\theta JSP}$) for UMC surface mount package.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

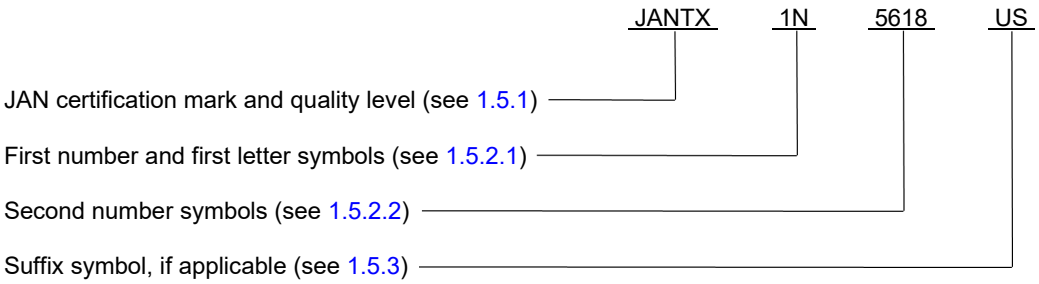
d. The complete PIN, see 1.5 and 6.5.

e. For die acquisition, the JANHC or JANKC letter version shall be specified (see figures 4 and 5).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

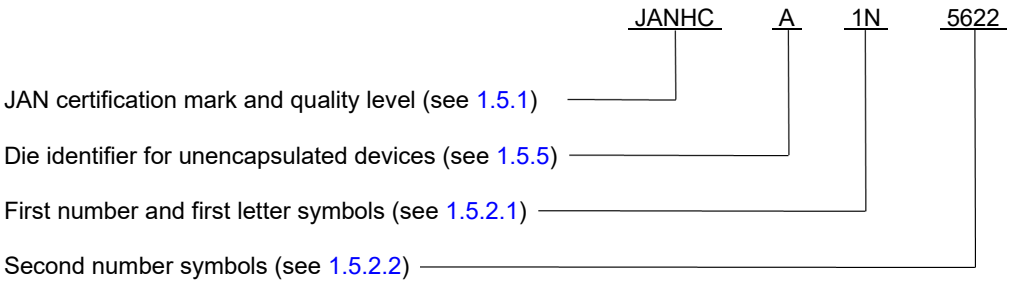
6.4 PIN construction example.

6.4.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.



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6.4.2 Un-encapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.



6.5 List of PINs.

6.5.1 List of PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

PINs for devices of the base quality level	PINs for devices of the "TX" quality level	PINs for devices of the "TXV" quality level	PINs for devices of the "S" quality level
Devices in an axial leaded package			
JAN1N5614	JANTX1N5614	JANTXV1N5614	JANS1N5614
JAN1N5616	JANTX1N5616	JANTXV1N5616	JANS1N5616
JAN1N5618	JANTX1N5618	JANTXV1N5618	JANS1N5618
JAN1N5620	JANTX1N5620	JANTXV1N5620	JANS1N5620
JAN1N5622	JANTX1N5622	JANTXV1N5622	JANS1N5622
Devices in a US package			
JAN1N5614US	JANTX1N5614US	JANTXV1N5614US	JANS1N5614US
JAN1N5616US	JANTX1N5616US	JANTXV1N5616US	JANS1N5616US
JAN1N5618US	JANTX1N5618US	JANTXV1N5618US	JANS1N5618US
JAN1N5620US	JANTX1N5620US	JANTXV1N5620US	JANS1N5620US
JAN1N5622US	JANTX1N5622US	JANTXV1N5622US	JANS1N5622US
Devices in a UMC package			
JAN1N5614UMC	JANTX1N5614UMC	JANTXV1N5614UMC	JANS1N5614UMC

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6.5.2 List of PINs for unencapsulated devices. The following is a list of possible PINs for unencapsulated devices available on this specification sheet. The qualified die suppliers with the applicable letter version (e.g., JANHCA1N5614) will be identified on the QML.

Type	Manufacturer	
	14552	34156
1N5614	JANHCA1N5614, JANKCA1N5614	JANHCB1N5614, JANKCB1N5614
1N5616	JANHCA1N5616, JANKCA1N5616	
1N5618	JANHCA1N5618, JANKCA1N5618	
1N5620	JANHCA1N5620, JANKCA1N5620	
1N5622	JANHCA1N5622, JANKCA1N5622	

6.6 Supersession data. The following MIL-PRF-19500/427 types supersede MIL-PRF-19500/228, MIL-PRF-19500/286, and MIL-S-19500/365 types, which are inactive for new design.

6.6.1 Superseded PINs. The following supersession data applies to PINs associated with this document:

Superseding type as specified within MIL-PRF-19500/427H, dated 8 July 2000	Superseded type as specified within MIL-S-19500/228G, 21 May 1993	Superseded type as specified within MIL-S-19500/286D, dated 15 March 1993	Superseded type as specified within MIL-S-19500/365A, NOTICE 1 28 December 1982
1N5614	1N3611	1N4245	1N4383
1N5616	1N3612	1N4246	1N4384
1N5618	1N3613	1N4247	1N4385
1N5620	1N3614	1N4248	1N4585
1N5622	1N3957	1N4249	1N4586

6.7 Applications data.

6.7.1 Half-sine-wave application with 1N5614 through 1N5622 (axial leaded and US). For a printed board mounting example with FR4 base material where the full 1 amp I_O rating (half-sine-wave) is used at a T_J = 175°C and ambient temperature of 55°C, the following steps guide the user in calculating what the printed board copper mounting pad size needs to be with 1 ounce, 2 ounce, and 3 ounce copper foil. For axial leaded, the lead length for mounting will be .187 inch (4.76 mm) or less from body to entry point on printed board surface.

a. Use the I_O versus P_O curve on figure 9 to look up 1 A (X-axis) and follow up to the T_J = 175°C curve (lower) for 0.90 watts.

b. Calculate maximum thermal resistance needed (175°C - 55°C) / 0.90W = 133 °C/W.

c. Locate the thermal resistance of 133°C/W on Y-axis using a thermal resistance versus copper mounting pad area plot on one of the three curves on figure 10 for different weights of copper foil and then intersect curve horizontally to get answer. These curves assume still air and horizontal printed board position.

d. In this example, the copper mounting pad sizes for the different copper foil weights would be as follows:

1) .030 in^2 (19.35 mm^2) for 1 ounce copper foil.
2) .018 in^2 (11.61 mm^2) for 2 ounce copper foil.
3) .011 in^2 (7.09 mm^2) for 3 ounce copper foil.

e. Add a conservative guard-band to the copper mounting pad size (larger) to keep T_J > 175°C.

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6.7.2 Square-wave application with 1N5614 through 1N5622 (axial leaded and US). For a PCB mounting example with FR4 base material to support a 0.5 amp I_O square wave switching at a 0.50 duty factor (50 percent duty cycle) at T_J = +125°C and ambient temperature of +55°C, the following steps guide the user in calculating what the PCB copper pad size needs to be with 1 ounce, 2 ounce, and 3 ounce copper foil.

a. Locate the size of copper pads on standard FR4 base material to support operation at 0.5 A I_O square wave switching at a 0.50 duty factor (50 percent duty cycle) at T_J = 125°C with T_A = 55°C.

b. Calculate peak I_F = 0.5A / 0.50 duty factor = 1 A.

c. Use the V_F versus I_F curve on figure 11 to look up I_F = 1 A (Y-axis) and follow across to the T_J = 125°C curve (middle) for V_F = 0.85 V.

d. Calculate power = $I_F * V_F *$ duty factor = $1 * 0.85 * 0.50 = 0.425$ W.

e. Calculate maximum thermal resistance needed (125°C - 55°C) / 0.425 W = 165 °C/W.

f. Locate the thermal resistance of 165 °C/W on the Y-axis using a thermal resistance versus copper pad area plot on one of the three curves on figure 10 for different weights of copper foil and then intersect curve horizontally to determine the answer. Curves assume still air and horizontal PCB position.

g. In this example, the copper pad sizes for the different copper foil weights would be as follows:

   1) .015 in^2 (9.67 mm^2) for 1 ounce copper foil.
   2) .009 in^2 (5.81 mm^2) for 2 ounce copper foil.
   3) .0061 in^2 (3.94 mm^2) for 3 ounce copper foil.

h. A conservative pad guard-band is optional since $T_J \geq 125$°C.

   NOTE: Multilayer PCBs, forced air cooling, etc. will improve performance. Closed confinement of the PCB will do the opposite. Please use sound thermal management.

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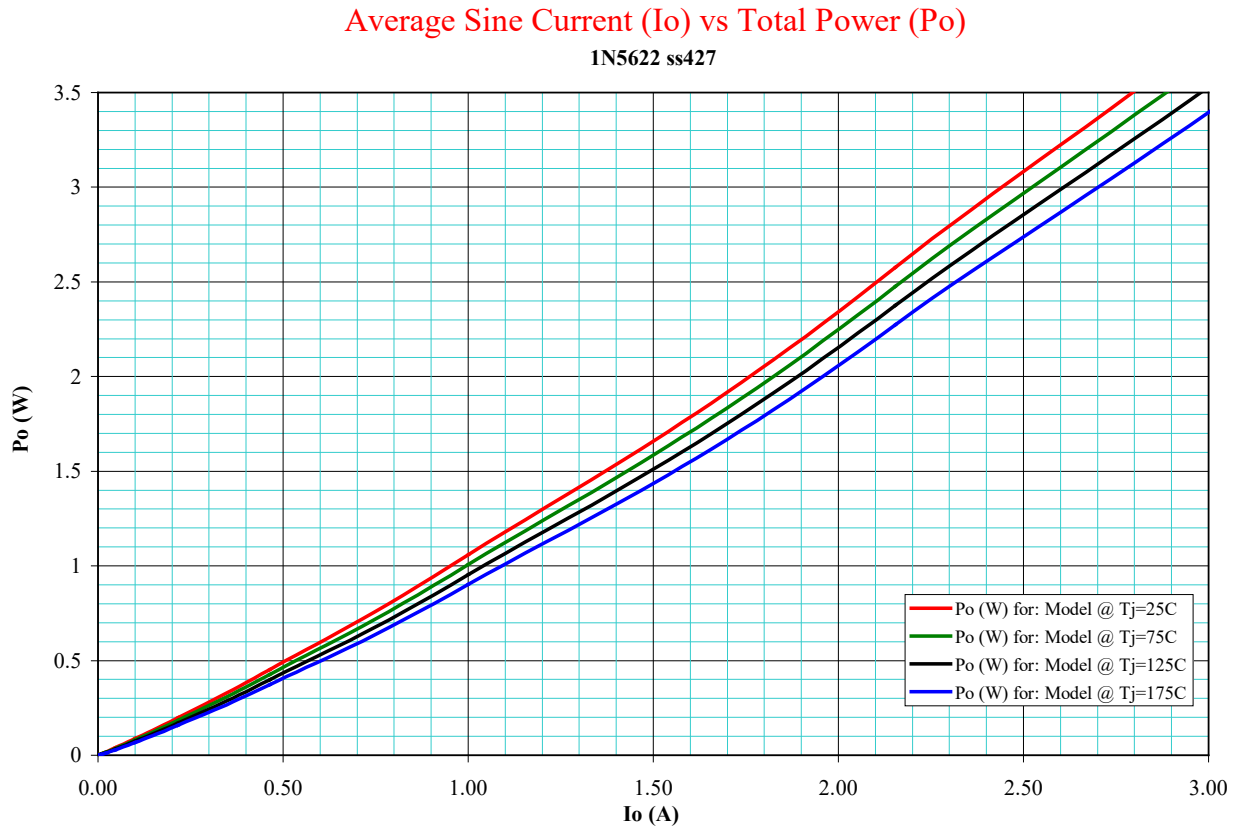
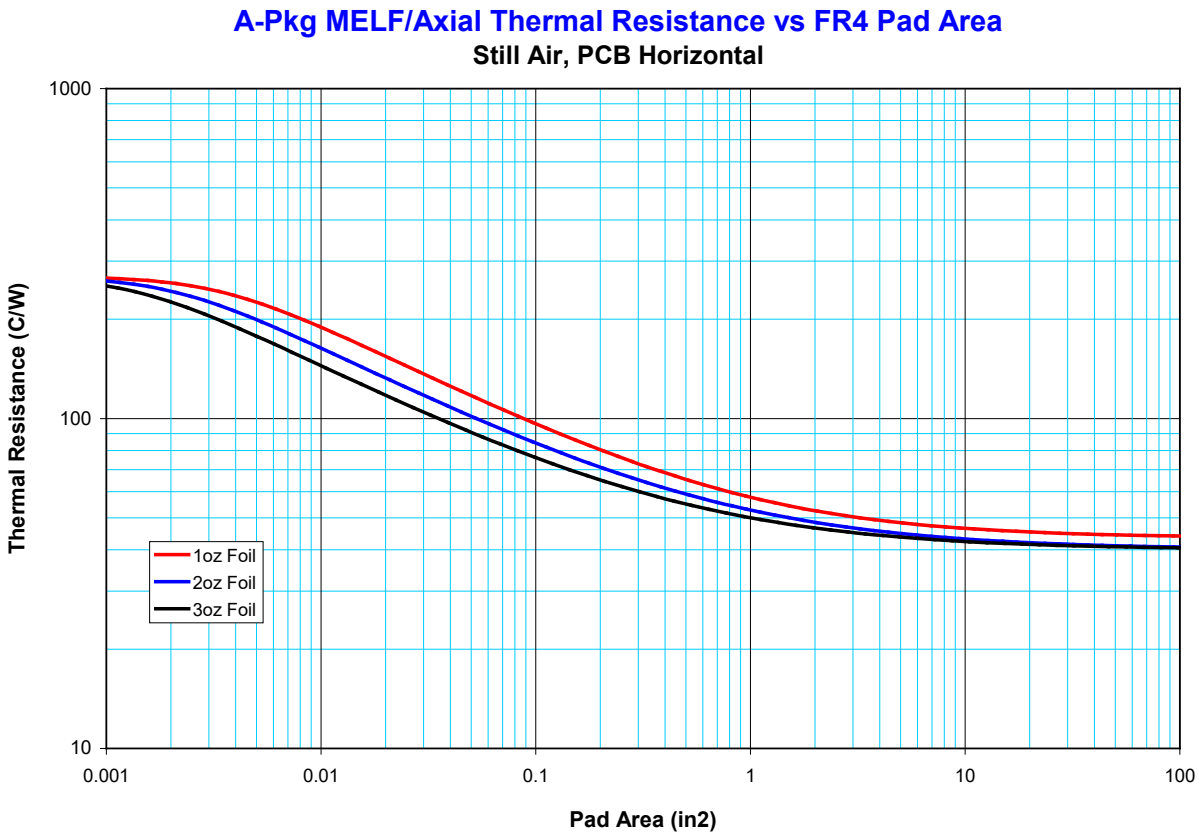


FIGURE 9. Rectifier power P_O versus I_O (average forward current) for 1N5614 through 1N5622 (axial leaded and US).

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NOTES:
1. PCB base material is FR4.
2. Copper foil weights are 1, 2, and 3 ounce per ft^2.
3. PCB mounting is horizontal in still air.

FIGURE 10. Thermal resistance versus pad area (for each pad) for 1N5614 through 1N5622 (axial leaded and US).

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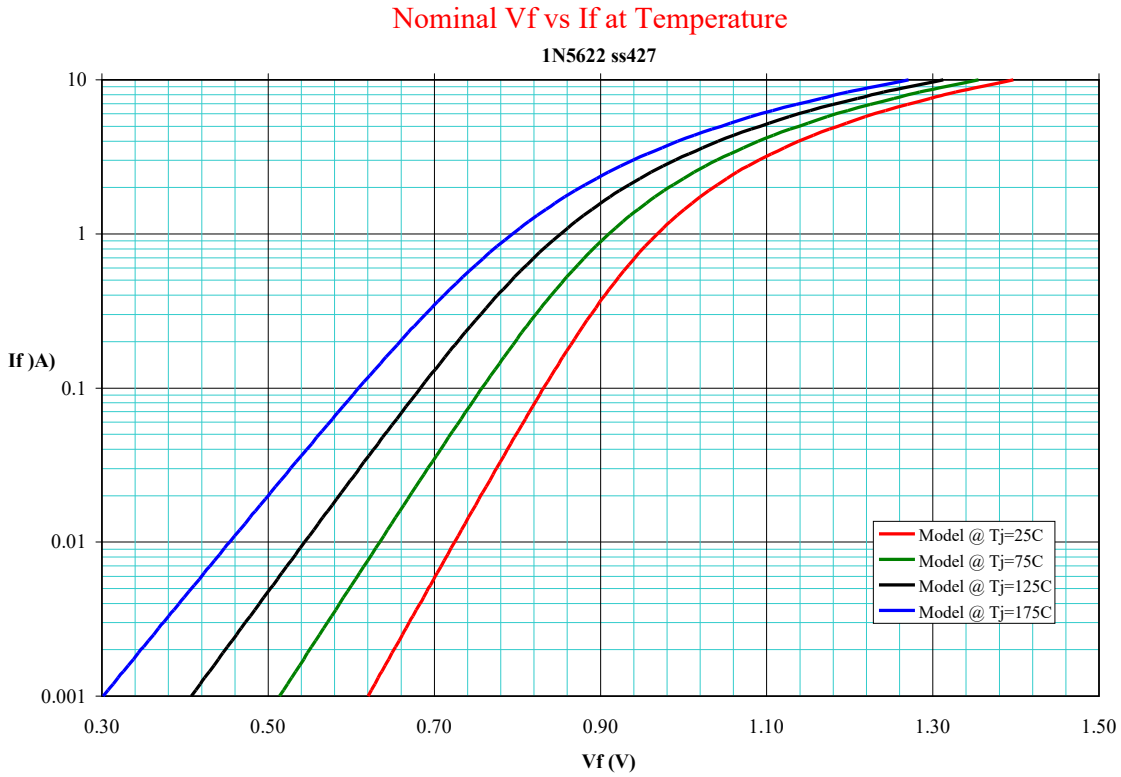


FIGURE 11. Forward voltage versus forward current for 1N5614 through 1N5622 (axial leaded and US).

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6.7.3 Half-sine-wave application with 1N5614UMC. For a PCB mounting example with FR4 base material there are two different applications to help bound specific mounting conditions. The first case is a FR4 PCB with different thicknesses of copper on the mounting side of the PCB only. The other case is a FR4 PCB with different thicknesses of copper on both sides of the PCB, this assumes a ground plane of the same copper thickness as the solder pads. See figure 12 for the relative difference in the solder pad sizes used in this example and the copper thickness for each pad are 1 oz, 2 oz, and 3 oz. The top image represents the actual size of the solder pads and shows the relative sizes for all the bond pads used in the examples. For the cases where there is copper on the back side of the PCB the thickness on the back will be equal to what is used for the solder pads.

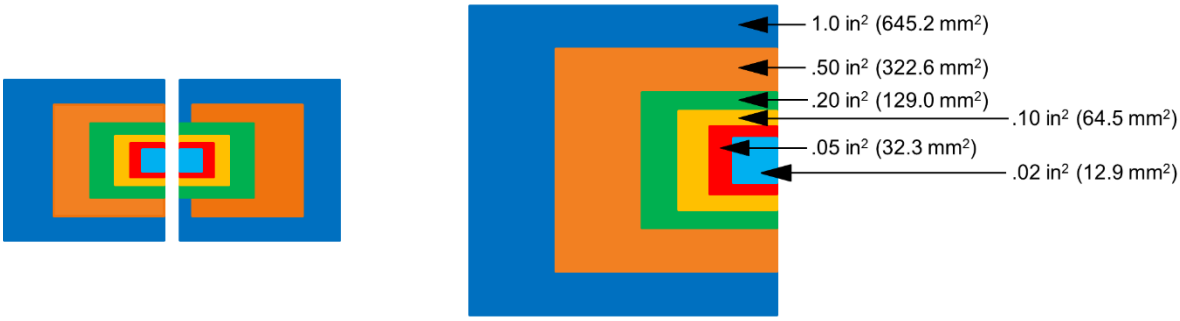


FIGURE 12. PCB and pad size used for examples.

It is assumed that the user will have set requirements for maximum junction temperature (T_J), maximum ambient temperature (T_A), and maximum operating current for the diode (I_O). The diode power (P_O) can be derived in figure 13 by finding the maximum operating current on the x-axis and using the y-axis curve that matches the required maximum junction temperature. Once the diode power is known the required thermal resistance can be calculated using the equation below.

$$R_{\theta JA} = \frac{T_J - T_A}{P}$$

The derived value for $R_{\theta JA}$ can then be used to define the size and thickness of the solder pad needed to match the operational requirement. Figure 14 can be used to size the solder pad for 1 oz, 2 oz, and 3 oz copper thicknesses and if the PCB has no copper ground plane or a copper ground plane of the same thickness.

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6.7.3.1 Example using T_J of 175°C. In the example where the operating conditions are 1 amp I_O (half-sine-wave), T_J of 175°C and ambient temperature of 55°C, the following steps can be used to determine the size of the solder pads for the 1N5614UMC.

a. Use the I_O versus P_O curve on figure 13 to look up 1 amp (X-axis) and follow up to the T_J =175°C curve (lower) for 0.94 watts.

b. Calculate maximum thermal resistance needed (175°C - 55°C) / 0.94 W = 127.7 °C/W.

c. Look up thermal resistance of 127.7 °C/W on Y-axis using a thermal resistance versus pad area plot on one of the three curves on figure 14 for different weights of copper cladding and then intersect curve horizontally to get the answer. These curves assume still air, horizontal position.

d. In this example, because the required thermal resistance is above any of the plots in figure 14, this implies any size of solder pad could be used and any thickness of copper could be used to meet the set of operational requirements.

e. Add a conservative guard-band to the pad size (larger) to keep T_J below 175°C.

6.7.3.2 Example using T_J of 125°C. In the example where the operating conditions are 1 amp I_O (half-sine-wave), T_J of 125°C and ambient temperature of 55°C, the following steps can be used to determine the size of the solder pads for the 1N5614UMC.

a. Use the I_O versus Po curve on figure 13 to look up 1 amp (X-axis) and follow up to the T_J =125°C curve (lower) for 1.0 watts.

b. Calculate maximum thermal resistance needed (125°C - 55°C) / 1.0 W = 70.0 °C/W.

c. Look up thermal resistance of 70.0 °C/W on Y-axis using a thermal resistance versus pad area plot on one of the three curves on figure 14 for different weights of copper cladding and then intersect curve horizontally to get the answer. These curves assume still air, horizontal position.

d. In this example, the copper mounting pad sizes for the different copper foil weights would be as follows:

1) .448 in^2 (289.0 mm^2) for 1 ounce copper foil.

2) .105 in^2 (67.7 mm^2) for 2 ounce copper foil.

3) .065 in^2 (3.94 mm^2) for 3 ounce copper foil.

These pad sizes are for each pad for a PCB without a ground plane and if a ground plane was employed, because the required thermal resistance is above any of the plots in figure 14, this implies any size of solder pad could be used and any thickness of copper could be used to meet the set of operational requirements.

e. Add a conservative guard-band to the pad size (larger) to keep T_J below 125°C.

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6.7.4 Square-wave application with 1N5614UMC. For a PCB mounting example with FR4 base material to support a 0.5 amp I_O square wave switching at a 0.50 duty factor (50 percent duty cycle) at T_J=125°C and ambient temperature of 55°C, the following steps guide the user in what the PCB pad size will need to be with 1 oz, 2 oz, and 3 oz copper foil, with and without a copper ground plane.

a. Find size of copper pads on standard FR4 PCB to support operation at 0.5 Amp I_O square wave switching at a 0.50 duty factor (50 percent duty cycle) at T_J = 125°C with T_A = 55°C.

b. Calculate peak I_F = 0.5A / 0.50 duty factor = 1 amp.

c. Use the V_F versus I_F curve on figure 15 to look up I_F = 1 A (Y-axis) and follow across to the T_J = 125°C curve for V_F = 0.736 V.

d. Calculate power = $I_F * V_F *$ duty factor = $1 * 0.736 * 0.50 = 0.368$W.

e. Calculate maximum thermal resistance needed (125°C - 55°C) / 0.368 W = 190.2 °C/W.

f. Look up thermal resistance of 190.2 °C/W on Y-axis using a thermal resistance versus pad area plot on one of the three curves on figure 14 for different weights of copper cladding and then intersect curve horizontally to get the answer. These curves assume still air, horizontal position.

g. In this example, because the required thermal resistance is above any of the plots in figure 14, this implies any size of solder pad could be used and any thickness of copper could be used to meet the set of operational requirements.

h. A conservative pad guard-band is optional since T_J is only 125°C.

NOTE: Multilayer PCBs, forced air cooling will improve performance. Closed confinement of the PCB will do the opposite. Use sound thermal management.

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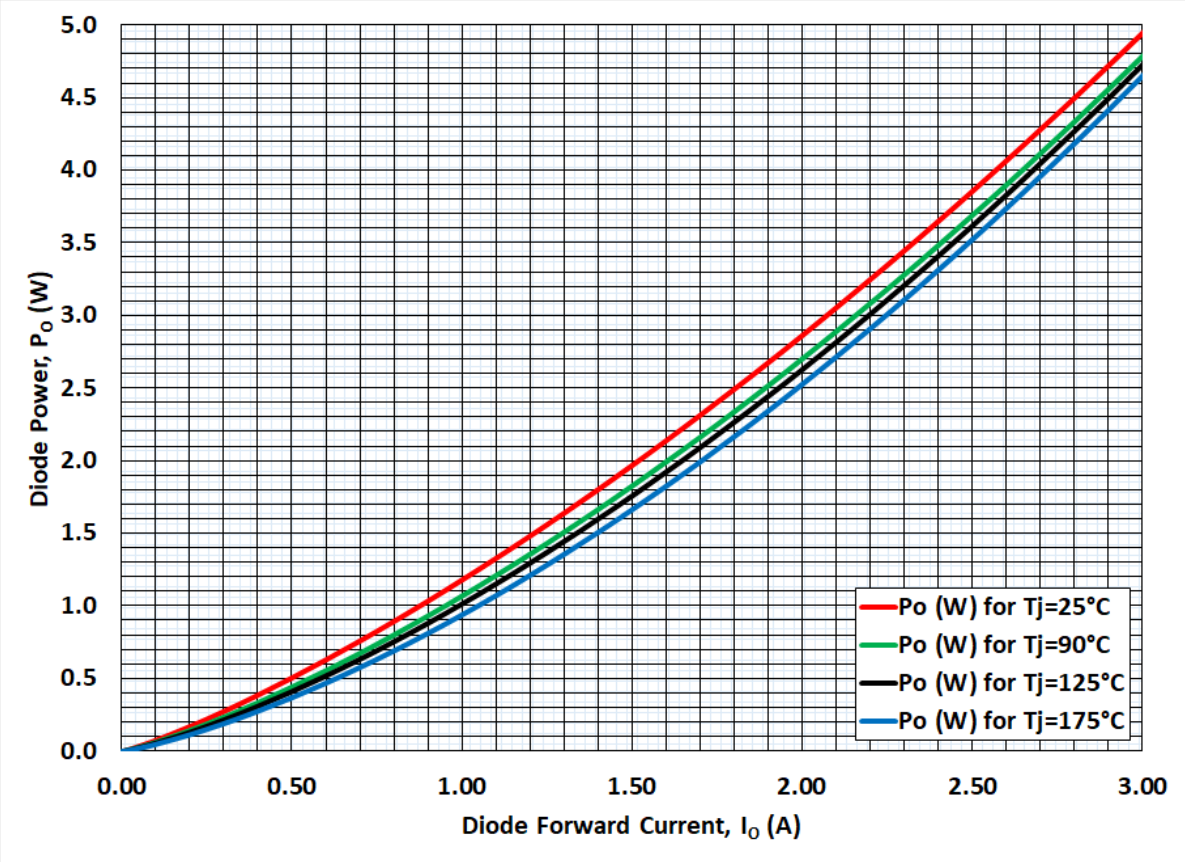
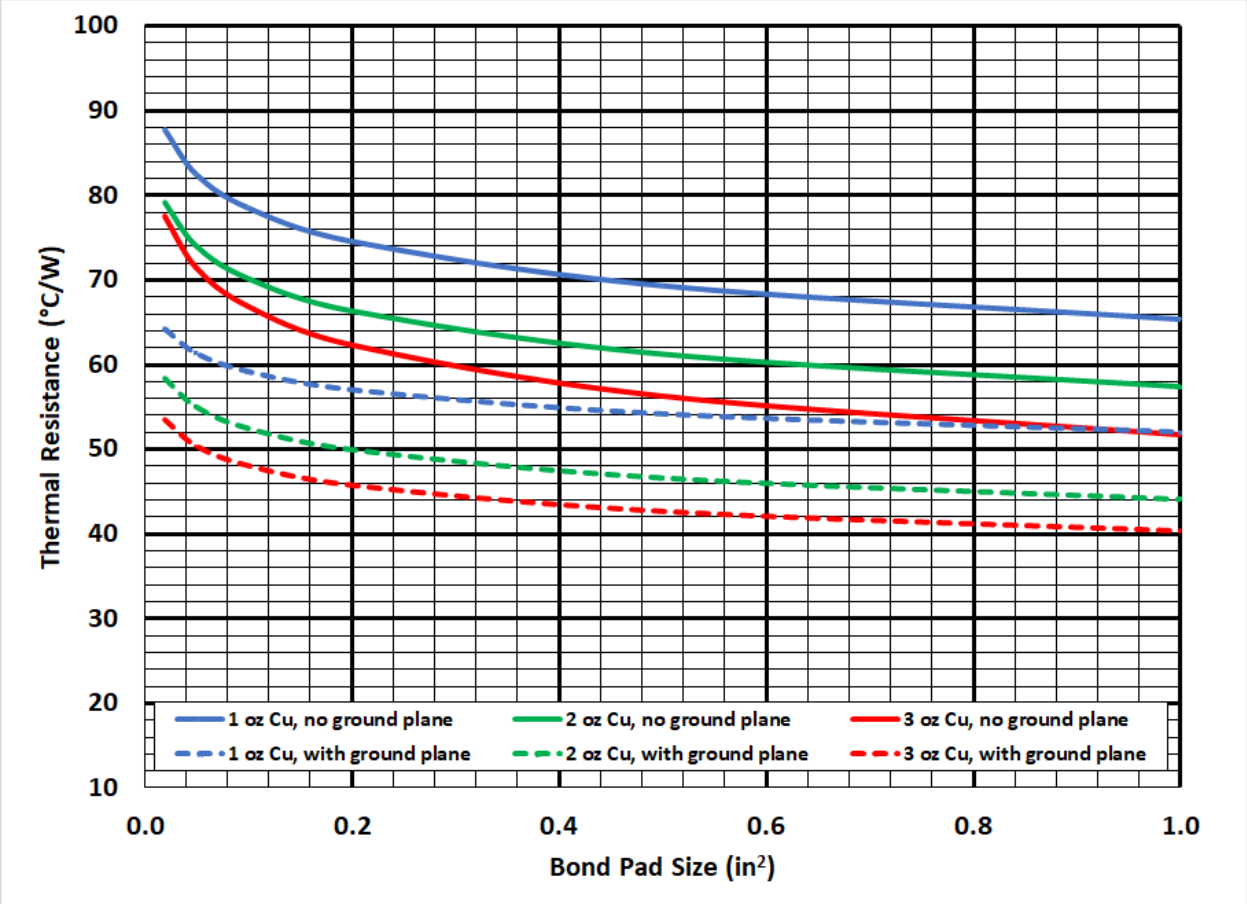


FIGURE 13. Diode P_O power versus I_O (average forward current) for 1N5614UMC.

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NOTES:
1. PCB base material is FR4 with 1, 2, and 3 oz copper, without or with ground plane.
2. PCB mounting is horizontal in still air.

FIGURE 14. Thermal resistance versus pad area (for each pad) for 1N5614UMC.

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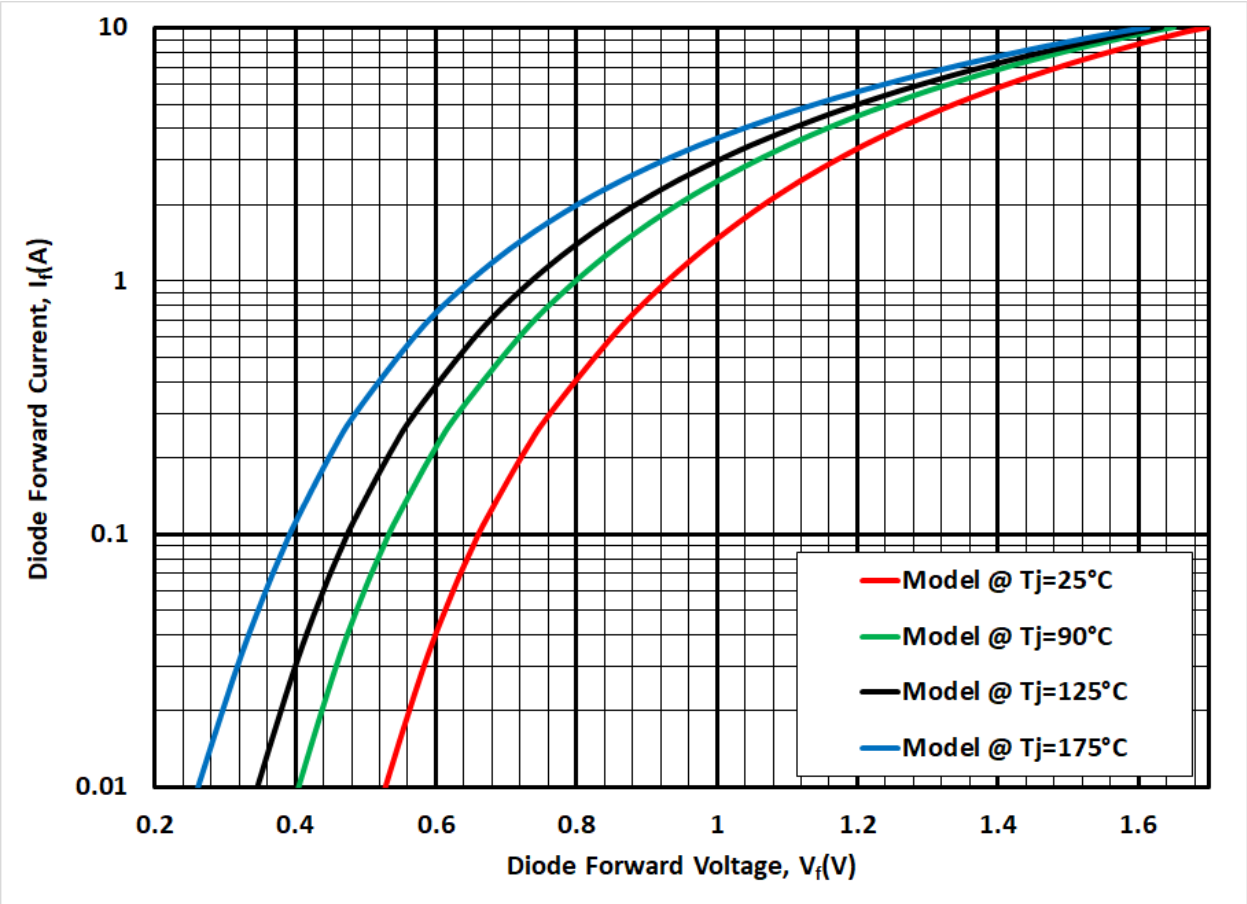


FIGURE 15. Forward voltage versus forward current for 1N5614UMC.

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6.8 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 692-6939 or DSN 850-6939.

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army − CR
Navy − SH
Air Force − 85
NASA − NA
DLA − CC

Review activities:
Army − AR, MI, SM
Navy − AS, MC
Air Force − 19

Preparing activity:
DLA − CC

(Project 5961-2022-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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